

JavaScript Assignment Questions

Topic-wise practice questions (minimum 5 per topic)

1. Variables & Data Types

- 1 Declare variables using var, let, and const, and explain the differences with examples.
- 2 Write a program to swap two numbers without using a third variable.
- 3 Explain the difference between null and undefined with code examples.
- 4 Write a program to check the data type of different values using typeof.
- 5 Convert a string to a number and a number to a string using different methods.

2. Operators

- 1 Write a program to demonstrate arithmetic, comparison, and logical operators.
- 2 Explain the difference between == and === with examples.
- 3 Write a program using the ternary operator to check if a number is even or odd.
- 4 Demonstrate the use of the nullish coalescing operator (??) with examples.
- 5 Write a program to calculate the area and perimeter of a rectangle using operators.

3. Conditional Statements

- 1 Write a program to check if a person is eligible to vote using if-else.
- 2 Write a program to find the largest of three numbers using if-else if.
- 3 Write a program to check whether a year is a leap year.
- 4 Write a program using switch-case to display the day of the week based on a number (1-7).
- 5 Write a program to categorize a student's grade based on marks (A, B, C, D, F).

4. Loops

- 1 Write a program to print the first 10 natural numbers using a for loop.
- 2 Write a program to calculate the factorial of a number using a while loop.
- 3 Write a program to print a multiplication table using a for loop.
- 4 Write a program to reverse a number using a loop.
- 5 Write a program to print the Fibonacci series up to n terms.

5. Functions

- 1 Write a function to check whether a number is prime or not.
- 2 Write a function to calculate the sum of digits of a number.
- 3 Explain the difference between function declarations and function expressions with examples.
- 4 Write an arrow function to find the square of a number.
- 5 Write a function with default parameters to calculate simple interest.

6. Arrays

- 1 Write a program to find the largest and smallest element in an array.
- 2 Write a program to remove duplicate elements from an array.
- 3 Demonstrate the use of `map()`, `filter()`, and `reduce()` with examples.
- 4 Write a program to sort an array of numbers in ascending and descending order.
- 5 Write a program to merge two arrays and remove duplicate values.

7. Strings

- 1 Write a program to check whether a string is a palindrome.
- 2 Write a program to count the number of vowels in a string.
- 3 Write a program to reverse a string without using built-in methods.
- 4 Write a program to check if two strings are anagrams of each other.
- 5 Write a program to capitalize the first letter of every word in a sentence.

8. Objects

- 1 Create an object to store student details (name, roll no, marks) and display them.
- 2 Write a program to loop through an object using `for...in`.
- 3 Write a program to merge two objects using the spread operator.
- 4 Write a program demonstrating object destructuring.
- 5 Write a program to check if a property exists in an object.

9. DOM Manipulation

- 1 Write a program to change the text content of an HTML element using JavaScript.
- 2 Write a program to add/remove a CSS class on button click.

- 3 Write a program to create a simple to-do list using DOM manipulation.
- 4 Write a program to validate a form using JavaScript.
- 5 Write a program to change the background color of a page on button click.

10. Error Handling

- 1 Explain try, catch, and finally with an example.
- 2 Write a program to handle division by zero using error handling.
- 3 Write a program that throws a custom error using throw.
- 4 Write a program to validate user input and handle invalid input using try-catch.
- 5 Explain the difference between syntax errors and runtime errors with examples.